

MOUNTAIN HOME AFB, ID, USA

WMO: 726815

Lat: 43.050N

Lon: 115.867W

Elev: 2996

StdP: 13.17

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24106

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	7.0	13.1	2.5	7.0	10.3	7.8	9.1	22.4	32.1	34.9	28.5	35.1	5.3	90	0.631

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	33.2	99.8	62.8	97.1	62.1	94.1	61.1	65.6	91.1	64.1	90.0	62.8	88.6	9.5	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.6	82.1	68.2	55.4	73.1	66.9	53.3	67.6	67.5	31.8	90.8	30.7	90.2	29.6	88.1	75.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.6	24.6	21.4	DB	0.6	104.8	7.4	2.5	-4.7	106.6	-9.1	108.1	-13.2	109.5	-18.6	111.3	(4)
			WB	-0.1	68.4	7.5	2.5	-5.5	70.2	-9.9	71.7	-14.1	73.1	-19.6	74.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	51.9	30.7	35.7	43.5	48.9	58.3	67.2	76.9	75.1	65.0	51.2	39.2	29.5	(6)
(7)		DBStd	17.68	8.43	7.20	6.47	7.22	7.70	7.85	5.70	5.74	7.41	7.89	8.69	8.74	(7)
(8)		HDD50	2392	598	400	219	106	18	1	0	0	2	78	334	636	(8)
(9)		HDD65	5739	1063	819	666	486	233	67	3	5	89	432	775	1101	(9)
(10)		CDD50	3070	0	1	18	71	274	515	835	779	452	116	9	0	(10)
(11)		CDD65	943	0	0	0	2	24	131	373	319	89	5	0	0	(11)
(12)		CDH74	14596	0	0	2	93	588	2207	5526	4543	1512	125	0	0	(12)
(13)		CDH80	7767	0	0	0	19	191	1065	3262	2554	647	29	0	0	(13)
(14)	Wind	WSAvg	9.7	9.4	10.8	10.9	11.1	10.2	10.0	9.2	8.7	8.3	9.0	9.4	9.8	(14)
(15)	Precipitation	PrecAvg	9.1	1.1	0.6	1.0	0.8	1.1	0.7	0.2	0.2	0.4	0.7	0.8	1.3	(15)
(16)		PrecMax	13.5	2.2	1.9	3.2	1.9	3.9	2.0	0.6	0.6	0.8	2.2	1.9	4.1	(16)
(17)		PrecMin	4.8	0.2	0.1	0.0	0.2	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.3	(17)
(18)		PrecStd	2.5	0.6	0.4	0.7	0.4	0.9	0.5	0.2	0.2	0.3	0.6	0.5	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.9	60.3	71.4	82.6	90.5	101.0	103.3	102.0	95.5	83.9	66.3	54.5	(19)
(20)		MCWB	43.7	48.4	50.1	55.9	59.1	64.1	64.1	63.4	61.5	57.2	50.5	46.1	(20)	
(21)		2%	DB	47.0	54.6	65.7	75.8	85.4	95.0	100.4	98.8	90.9	77.5	60.8	48.9	(21)
(22)		MCWB	41.4	44.6	48.2	53.0	57.5	61.9	63.0	62.7	59.8	54.1	47.9	43.5	(22)	
(23)		5%	DB	43.9	50.4	61.2	70.3	80.9	90.6	98.3	95.9	87.7	71.9	55.8	45.1	(23)
(24)		MCWB	39.2	42.3	46.7	50.6	56.0	60.1	62.3	61.4	58.6	52.4	45.6	40.9	(24)	
(25)		10%	DB	41.0	46.8	57.0	64.4	75.9	86.4	95.2	93.0	83.6	66.3	52.1	41.4	(25)
(26)		MCWB	37.6	40.9	44.8	48.5	54.5	58.9	61.7	60.8	57.5	50.2	44.5	38.2	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.4	49.4	53.0	57.3	61.0	66.2	68.4	67.6	64.3	59.0	53.4	48.3	(27)
(28)		MCDWB	48.7	57.1	61.4	77.9	82.9	93.0	92.4	92.3	85.9	73.9	60.5	51.9	(28)	
(29)		2%	WB	42.6	46.4	50.6	54.2	59.2	63.7	65.9	65.1	61.9	56.5	49.9	44.4	(29)
(30)		MCDWB	45.8	52.4	60.7	72.4	79.8	90.4	92.6	88.7	84.1	72.2	57.8	47.9	(30)	
(31)		5%	WB	40.3	43.8	48.5	51.8	57.4	61.8	64.4	63.5	60.1	53.9	47.3	41.5	(31)
(32)		MCDWB	43.0	48.9	58.5	66.8	76.3	85.6	91.3	89.2	82.5	68.4	54.4	44.5	(32)	
(33)		10%	WB	38.2	41.4	46.3	49.8	55.7	60.0	63.2	62.2	58.4	51.6	44.9	38.6	(33)
(34)		MCDWB	40.4	46.1	55.9	62.8	72.5	81.6	89.9	87.9	80.3	64.4	50.9	41.4	(34)	
(35)	Mean Daily Temperature Range	MDBR	14.1	17.8	23.2	25.8	27.4	29.9	33.2	33.2	31.6	26.8	20.1	14.5	(35)	
(36)		5% DB	MCDBR	16.7	21.7	29.3	35.6	35.3	37.6	37.3	37.1	37.0	33.0	23.9	17.1	(36)
(37)		MCWBR	11.5	13.2	15.6	16.3	14.0	14.0	13.0	13.3	14.7	15.0	13.4	11.6	(37)	
(38)		5% WB	MCDBR	15.3	19.5	25.0	30.4	30.8	33.3	32.4	32.8	31.8	28.2	22.0	14.9	(38)
(39)		MCWBR	11.3	12.8	14.1	14.6	12.7	12.6	12.3	12.3	12.7	13.6	13.0	10.9	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.260	0.268	0.283	0.314	0.327	0.331	0.341	0.341	0.314	0.287	0.269	0.262	(40)	
(41)		taud	2.578	2.591	2.572	2.465	2.462	2.479	2.463	2.447	2.514	2.584	2.587	2.570	(41)	
(42)		Ebn at Noon	287	302	308	303	299	298	293	290	292	290	281	276	(42)	
(43)		Edh at Noon	20	24	28	33	35	34	34	34	29	24	20	19	(43)	
(44)	All-Sky Solar Radiation	RadAvg	547	850	1254	1725	2087	2406	2434	2106	1660	1105	658	443	(44)	
(45)		RadStd	69	114	123	134	170	146	112	104	93	77	52	66	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
(47)	Regional (12 neighbors)	N/A	-2.30	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page